ECA18 - OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO2 PROBLEM SOLVING ASSIGNMENT REPORT

Standards-Compliant Laser Transmitter Design, Wavelength Control and Nonlinearity Management for an 8-Channel DWDM Link

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Executive Summary & Design Requirement Statement

This engineering report develops a complete transmitter solution for an 8-channel DWDM fiber-optic link operating around the 1550 nm window. Each channel carries 25 Gb/s NRZ data with 100 GHz channel spacing over six amplified 80 km spans of standard single-mode fiber. The design must provide adequate optical power, high modulation speed, narrow spectral linewidth, stable wavelength under a 40 °C ambient variation, controlled nonlinear penalties, eye-safety, and a practical implementation path.

The supplied laser data give an emission wavelength of 1.55 µm from E_g = 0.80 eV and an internal quantum efficiency of 80%. At the specified 40 mA bias, the calculated output is 8.4 mW (9.24 dBm). The 25 Gb/s NRZ bandwidth requirement is 17.5 GHz, leaving a 2.5 GHz margin below the stated 20 GHz modulation bandwidth. However, the temperature-induced wavelength drift would be 4.0 nm, five times the 0.8 nm channel spacing, so active thermal stabilization is essential.

For nonlinear transmission, the 80 km effective length is approximately 21.17 km after converting 0.2 dB/km attenuation to nepers/km. The SPM phase shift is approximately 0.055 rad per span and 0.330 rad over six spans. With the stated scattering relations and conventional silica coefficients assumed as g_B = 5×10^-11 m/W and g_R = 6×10^-14 m/W, the SBS threshold is about 1.59 mW and the SRS threshold about 1.01 W. Therefore SBS is the principal threshold concern at the 2 mW/channel launch level, while SRS is far above the total 16 mW launch power.

The 15 ps pulse has a dispersion length of 11.25 km, whereas the nonlinear length at 2 mW is 384.6 km. The resulting soliton order N = 0.171 is well below unity; therefore the specified launch power does not support fundamental soliton transmission. Digital backpropagation is more appropriate as a receiver/DSP compensation method, although full-link multi-channel implementation is computationally intensive. A Si-photonics transmitter with a hybrid-integrated narrow-linewidth laser is recommended for scalable 8-channel implementation, with external modulation, TEC wavelength stabilization, optical filtering/isolation, and DSP-assisted nonlinear compensation.

Problem Requirement Statement

The proposed 8-channel DWDM transmitter shall provide eight 25 Gb/s NRZ channels around 1550 nm with 100 GHz spacing, sufficient optical output power, at least 20 GHz small-signal modulation bandwidth, narrow spectral width and high side-mode suppression, wavelength stability within ±0.1 nm, controlled chirp, and safe operation. The architecture shall limit SPM/XPM/FWM/SBS/SRS penalties, support six 80 km amplified spans, and provide a practical integration and DSP strategy consistent with the stated standards references.

Stage 1: Interpretation of the Communication Requirement

Working backward from the 8-channel, 25 Gb/s DWDM requirement, the transmitter must simultaneously satisfy speed, spectral, power, thermal, and nonlinear constraints.

- High modulation speed is required because each channel carries 25 Gb/s NRZ data. The given relation B ≈ 0.7R_b gives a minimum bandwidth of 17.5 GHz.
- Narrow linewidth and high side-mode suppression are important because adjacent DWDM channels are separated by only 100 GHz. Excess spectral spreading, chirp and mode competition increase crosstalk and reduce receiver margin.
- Output power must be sufficient for six 80 km spans, while excessive launch power increases Kerr nonlinearities and stimulated scattering. The specified 2 mW/channel launch gives 16 mW total for eight channels.
- Laser threshold current, slope efficiency, quantum efficiency and bandgap determine whether the device can produce useful optical power efficiently. Linewidth and side-mode suppression determine spectral purity.
- Direct modulation changes carrier density and refractive index, producing frequency chirp. External modulation separates intensity modulation from the laser cavity and is therefore preferred when dispersion and WDM spectral stability are important.
- Temperature control is critical because the supplied wavelength coefficient is 0.1 nm/°C. Without stabilization, a 40 °C ambient change produces a 4 nm shift.
- SPM, XPM and FWM arise from Kerr nonlinearity; SBS and SRS arise from stimulated scattering. Their importance depends on launch power, channel spacing, dispersion, effective length and the number of spans.

Standards role: ITU-T G.957 is relevant as an optical-interface reference for applicable optical transmission interfaces; IEEE 802.3bs/802.3bm provide Ethernet PHY/interface considerations for 100G/200G/400G-class systems; and ANSI Z136 is used for laser eye-safety assessment. These references should be treated as compliance/compatibility checks rather than as a single governing specification for this exact 8×25 Gb/s DWDM link.

Stage 2: Investigation and Selection of the Optical Source / Transmitter Configuration

Bandgap, wavelength and internal quantum efficiency

The photon energy is related to wavelength by λ(μm) ≈ 1.24/E_g(eV). Therefore:

λ = 1.24 / 0.80 = 1.55 μm = 1550 nm.

The internal quantum efficiency is η_i = τ_nr/(τ_r + τ_nr) = 40/(10 + 40) = 0.80 = 80%. The result indicates that 80% of recombination events contribute through the radiative channel under the supplied lifetime model.

Candidate source comparison

Source	Output power	Spectral width / purity	Modulation speed	Efficiency / threshold	DWDM suitability
LED	Low	Broad spectrum	Low	High spontaneous emission; no laser threshold	Poor
FP laser	Moderate–high	Multiple longitudinal modes	High	Moderate efficiency; threshold operation	Limited by multimode spectrum

DBR laser	High	Narrow; wavelength-selective	High	Good efficiency; tunable/controlled	Good
DFB laser	High	Very narrow; high SMSR	High	Low threshold, high slope efficiency	Excellent
VCSEL	Low–moderate	Narrow	Very high	Low threshold; excellent modulation	Good, but 1550 nm long-haul versions are specialized
QD laser	Moderate–high	Narrow potential	High	Low threshold potential; temperature tolerant	Promising, but less conventional for this link

The DFB laser is selected because it combines narrow linewidth, high side-mode suppression, high modulation speed, suitable 1550 nm operation, good fiber coupling and high efficiency. A DBR laser is a credible alternative when active wavelength tuning across the DWDM grid is prioritized.

Laser oscillation, linewidth and side-mode suppression

Laser oscillation begins when the modal gain reaches the total cavity loss (threshold condition: $\Gamma g_{th} \approx \alpha_i + \alpha_m$). Below threshold, spontaneous emission dominates; above threshold, stimulated emission produces coherent optical power. A narrow linewidth reduces spectral overlap and phase noise, while high side-mode suppression ratio (SMSR) suppresses unwanted longitudinal modes. Both are important in an 8-channel, 100 GHz-spaced DWDM system.

DESIGN DECISION 1 — OPTICAL SOURCE: I recommend a 1550 nm DFB laser optical source because it provides the best combination of narrow spectral width, high SMSR, high-speed modulation capability, efficient optical output and strong single-mode fiber coupling for the six-span DWDM link.

A DBR laser becomes preferable when wide, fast wavelength tuning is more important than the simplest high-volume fixed-grid implementation.

Stage 3: Modulation, Chirp and Spectral-Stability Strategy

Output power

Given $P = \eta_s(I - I_{th})$:

$P = 0.3\ \text{mW/mA} \times (40 - 12)\ \text{mA} = 8.4\ \text{mW}$.

$P(\text{dBm}) = 10 \log_{10}(8.4) = 9.24\ \text{dBm}$.

Thus the specified bias produces 8.4 mW optical output, which is above the 2 mW/channel launch level. A variable optical attenuator or splitter/coupler can set the actual launch level while avoiding unnecessary nonlinear penalties.

Modulation bandwidth

For $R_b = 25$ Gb/s, $B_{req} \approx 0.7R_b = 0.7 \times 25 = 17.5$ GHz. Given $B_{3dB} = 20$ GHz, bandwidth margin = $20 - 17.5 = 2.5$ GHz, or about 14.3% relative to the required bandwidth. The source is therefore adequate, but the margin is modest and leaves limited allowance for packaging, driver, parasitic and temperature effects.

Carrier and photon rate equations

A standard semiconductor-laser rate-equation representation is:

$$dN/dt = I/(qV) - N/\tau_n - \Gamma v_g g(N)S$$

$$dS/dt = \Gamma v_g g(N)S + \beta_{sp} N/\tau_{sp} - S/\tau_p$$

where N is carrier density, S is photon density, I is injection current, q is electron charge, V is active volume, Γ is confinement factor, v_g is group velocity, g(N) is material gain, τ_n is carrier lifetime, τ_p is photon lifetime, and β_{sp} represents spontaneous-emission coupling.

The relaxation-oscillation frequency is the characteristic carrier-photon resonance. In small-signal modulation it determines the frequency response and peaking/bandwidth; in large-signal modulation the carrier and photon populations undergo nonlinear transients, so turn-on delay, pattern dependence, damping and chirp become important.

Direct versus external modulation

Aspect	Direct modulation	External modulation
Chirp	Higher because carrier-density changes alter refractive index	Lower intensity-to-frequency coupling; can be designed for low chirp
Bandwidth	Limited by laser relaxation dynamics and parasitics	Can exceed laser direct-modulation limits depending on modulator
Dispersion interaction	More sensitive because chirp converts dispersion into pulse distortion	Better for long-haul/high-speed transmission
Complexity	Low	Higher
Recommended here	Not preferred	Preferred

For six 80 km spans and tightly spaced DWDM channels, the reduction of chirp and improved spectral control outweigh the added complexity of external modulation. An integrated or hybrid-integrated Mach–Zehnder/electro-optic modulator is therefore selected.

Wavelength drift and temperature control

Uncontrolled wavelength drift = 0.1 nm/°C × 40 °C = 4.0 nm. The DWDM channel spacing is approximately 0.8 nm, so the uncontrolled drift is 4.0/0.8 = 5 channel spacings. This is unacceptable.

For ±0.1 nm tolerance, the allowed temperature variation is ±0.1/0.1 = ±1 °C, i.e. a total controlled window of about 2 °C. A TEC should therefore maintain the laser temperature to approximately ±1 °C or tighter. Injection-current control can provide fine tuning and fast correction, but it also changes carrier density and can introduce chirp and power variation. TEC control is preferred for primary wavelength stabilization, with current tuning used only as a fine-control/tracking mechanism.

DESIGN DECISION 2 — MODULATION CONFIGURATION: I recommend external modulation and TEC-based wavelength control because the calculated 17.5 GHz requirement leaves only 2.5 GHz bandwidth margin, while uncontrolled thermal drift is 4.0 nm. External modulation reduces chirp/dispersion interaction and TEC stabilization holds the wavelength within ±0.1 nm.

Stage 4: Optical Nonlinearities and Control

Effective length and SPM

The supplied attenuation is 0.2 dB/km. For the exponential effective-length equation, α must be expressed in nepers/km:

$\alpha = 0.2 \times \ln(10)/10 = 0.04605\ \text{km}^{-1}$.

$L_{eff} = (1 - e^{(-\alpha L)})/\alpha = (1 - e^{(-0.04605\times80)})/0.04605 \approx 21.17$ km.

For P = 2 mW = 0.002 W and $\gamma = 1.3\ \text{W}^{-1}\text{km}^{-1}$:

$\phi_{NL} = \gamma P L_{eff} = 1.3\times0.002\times21.17 \approx 0.055$ rad per span.

For six spans: $\phi_{NL,total} \approx 6\times0.055 = 0.330$ rad. This phase shift is not negligible and increases with launch power.

XPM and FWM

- XPM is produced because the intensity of neighboring DWDM channels changes the refractive index seen by a channel through the Kerr effect. It causes phase modulation and can become important as channel count and launch power increase.
- FWM generates new frequencies from combinations of channel frequencies. With 100 GHz spacing, generated products can fall close to existing channels, causing crosstalk. Lower chromatic dispersion generally increases phase matching and can worsen FWM.
- Both XPM and FWM become more severe with higher launch power. Dispersion, channel spacing, span length and channel power must therefore be considered together.

SBS and SRS threshold calculations

The assignment gives P_th(SBS) ≈ 21A_eff/(g_B L_eff) and P_th(SRS) ≈ 16A_eff/(g_R L_eff). For the calculations below, the standard silica-order coefficients are taken as g_B = 5×10^-11 m/W and g_R = 6×10^-14 m/W because the exponent signs in the supplied parsed statement are not visually explicit. A_eff = 80 μm² = 80×10^-12 m² and L_eff = 21.17 km = 2.117×10^4 m.

SBS threshold = 21(80×10^-12)/(5×10^-11×2.117×10^4) ≈ 1.59×10^-3 W = 1.59 mW.

SRS threshold = 16(80×10^-12)/(6×10^-14×2.117×10^4) ≈ 1.01 W = 1008 mW.

Comparison: the 2 mW/channel launch is slightly above the calculated SBS threshold, whereas the total 16 mW launch is far below the calculated SRS threshold. Hence SBS is the dominant stimulated-scattering concern. Narrow linewidth increases SBS susceptibility because a narrow optical spectrum concentrates power within the Brillouin gain bandwidth; deliberate linewidth broadening/dithering can raise the effective SBS threshold when needed.

Eye safety and nonlinear mitigation

The total launch power is 8×2 mW = 16 mW, corresponding to 12.04 dBm. For assignment-level screening, an assumed accessible-power ceiling of 100 mW (20 dBm) is used; 16 mW is below this assumed limit. The actual ANSI Z136 classification must be verified from the complete accessible-emission, wavelength, exposure-time, connector and enclosure conditions rather than total transmitter power alone.

Two practical ways to reduce nonlinear penalties without unnecessarily reducing received signal are: (1) use per-channel power optimization/power equalization so no channel is over-launched, and (2) use dispersion management and/or DSP-based nonlinear compensation. For SBS specifically, controlled phase modulation/linewidth broadening can also increase the SBS threshold.

DESIGN DECISION 3 — NONLINEAR-CONTROL STRATEGY: I recommend optimized per-channel launch power, optical filtering/equalization, SBS-aware linewidth/phase dithering, and receiver-side DSP compensation because SBS is near the 2 mW/channel level while SPM/XPM/FWM accumulate across six spans.

Stage 5: Soliton Transmission, Digital Backpropagation and Photonic Integration

Soliton condition

Soliton transmission requires a balance between anomalous chromatic dispersion and Kerr self-phase modulation. The relevant condition is L_D ≈ L_NL for a fundamental soliton.

Given T = 15 ps and $|\beta_2|$ = 20 ps²/km:

$L_D = T^2/|\beta_2| = 15^2/20 = 11.25$ km.

At 2 mW/channel: $L_{NL} = 1/(\gamma P) = 1/(1.3 \times 0.002) = 384.6$ km.

Soliton order: $N = \sqrt{(L_D/L_{NL})} = \sqrt{(11.25/384.6)} = 0.171$.

For a fundamental soliton, N = 1, so the required peak power is $P_1 = 1/(\gamma L_D) = 1/(1.3 \times 11.25) = 0.0684$ W ≈ 68.4 mW. The specified 2 mW/channel is therefore much too low for a fundamental soliton. Soliton transmission is NOT feasible for the specified pulse and launch power.

Digital backpropagation

Digital backpropagation numerically applies an inverse propagation model to compensate accumulated chromatic dispersion and Kerr nonlinear phase effects. It can jointly address dispersive and nonlinear impairments more effectively than treating each effect independently. A practical limitation is high computational complexity, especially when multiple DWDM channels and six spans are modeled jointly. For this reason, full-bandwidth 8-channel DBP is not the simplest transmitter-only solution. A practical architecture uses moderate-complexity per-channel or reduced-bandwidth DSP, with optical power optimization and filtering before the receiver.

Standards and implementation strategy

Reference / technology	Role in the design	Verification approach
ITU-T G.957	Optical interface reference for applicable high-speed optical transmission interfaces	Check wavelength, launch/receive optical parameters and interface limits where the selected interface profile applies
IEEE 802.3bs / 802.3bm	100G/200G/400G Ethernet PHY/interface reference points	Check relevant optical PMD/PHY assumptions if Ethernet client interfaces are used
ANSI Z136	Laser safety framework	Verify accessible emission and exposure classification for the actual package/system
Si-photonics + hybrid laser	Scalable multi-channel transmitter platform	Verify optical coupling, thermal control, modulator loss, crosstalk and reliability

Implementation	Footprint	Power	Coupling / reliability	Scalability	Assessment
Discrete components	Large	Higher interconnect/driver overhead	Flexible but more interfaces	Moderate	Good for prototype; less attractive for 8 channels
Monolithic InP	Compact	Potentially low	Excellent integration of active optical functions	High	Strong option when full active integration is prioritized
Si-photonics + hybrid DFB laser	Very compact	Low-to-moderate system overhead	Efficient passive routing; hybrid laser adds coupling/thermal design	Excellent	Recommended for scalable 8-channel transmitter

DESIGN DECISION 4 — INTEGRATED TRANSMITTER: I recommend a Si-photonics platform with a hybrid-integrated 1550 nm DFB laser because it provides compact wavelength routing, scalable multi-channel integration, reduced interconnect count and a practical path to integrating modulators, multiplexers, monitors and control electronics. DSP-based nonlinear compensation should be placed in the receiver/DSP chain rather than relying on soliton transmission.

Final DWDM Transmitter Block Diagram

Proposed architecture:

Block 1	Block 2	Block 3	Block 4	Block 5	Block 6	Block 7
8 × 1550 nm DFB lasers	TEC + wavelength lock / monitor	External MZ modulators	8-channel DWDM MUX	Optical power control + filter	EDFA / span launch interface	6 × 80 km SMF spans → RX/DSP

Control loops: wavelength monitor → TEC/current controller; optical monitor → power equalization. At the receiver, coherent/direct-detection DSP as appropriate performs dispersion and nonlinear compensation. Optical isolation and filtering are included to suppress reflections and adjacent-channel leakage.

Final Parameter / Design Table

Parameter	Adopted value / result	Design justification
Number of channels	8	Required DWDM configuration
Channel spacing	100 GHz ≈ 0.8 nm	Given requirement
Data rate	25 Gb/s NRZ/channel	Given requirement
Operating wavelength	1550 nm	From E_g = 0.80 eV; low-loss fiber window
Optical source	DFB laser	Narrow linewidth, high SMSR, high-speed DWDM suitability
Internal quantum efficiency	80%	$\tau_{nr}/(\tau_r+\tau_{nr})$ = 40/50
Bias current	40 mA	Given
Threshold current	12 mA	Given
Calculated laser output	8.4 mW = 9.24 dBm	$\eta_s(I-I_{th})$
Actual launch target	2 mW/channel = 3.01 dBm	Given launch condition; attenuate/equalize from available output
Modulation bandwidth	20 GHz	Given source data
Required NRZ bandwidth	17.5 GHz	0.7×25 Gb/s
Bandwidth margin	2.5 GHz	20−17.5 GHz
Modulation	External MZ/electro-optic modulation	Lower chirp and better long-haul DWDM behaviour
Wavelength control	TEC primary + fine current tuning	±0.1 nm requires about ±1 °C thermal control
Uncontrolled thermal drift	4.0 nm	0.1 nm/°C × 40 °C
Effective length per 80 km span	21.17 km	α = 0.2 dB/km converted to 0.04605 km^{-1}
SPM phase shift	0.055 rad/span; 0.330 rad/6 spans	At 2 mW/channel
SBS threshold	≈1.59 mW	Below/near 2 mW channel launch; dominant scattering concern
SRS threshold	≈1.01 W	Far above 16 mW total launch
Dispersion length	11.25 km	$15^2/20$
Nonlinear length	384.6 km	1/(1.3×0.002)
Soliton order	N = 0.171	Well below 1
Fundamental soliton power	≈68.4 mW peak	$1/(\gamma L_D)$; not available at 2 mW launch
Soliton feasibility	NO	N ≠ 1 and launch power too low
Nonlinear compensation	Reduced-complexity DBP / DSP + power optimization	Practical alternative to soliton transmission
Integration	Si-photonics + hybrid DFB laser	Scalable 8-channel implementation
Total launch power	16 mW = 12.04 dBm	8×2 mW
Eye-safety assumption	20 dBm screening ceiling; actual ANSI Z136 check required	16 mW is below assumed screening level

Stage 6: Verification Table and Final System Design

Requirement	Specified / Target	Achieved Value	Satisfactory?
Optical source selection	High-speed, narrow linewidth, high power, stable single-mode output	1550 nm DFB; 8.4 mW calculated output; high-SMSR architecture	YES
Modulation bandwidth and output power	B ≥ 17.5 GHz; suitable launch power	20 GHz; 2.5 GHz margin; 8.4 mW available, 2 mW/channel launch controlled	YES
Spectral/chirp and wavelength stability	±0.1 nm; low chirp	External modulation; TEC maintains approximately ±1 °C; fine current tuning	YES
Nonlinear-impairment control and eye safety	Control SPM/XPM/FWM/SBS/SRS; safe operation	SPM 0.330 rad total; SBS identified as dominant; power equalization/linewidth control/DSP; 16 mW total	YES*
Soliton / dispersion condition	Fundamental soliton if used	L_D = 11.25 km; L_NL = 384.6 km; N = 0.171; soliton not used	YES – alternative selected
Integration / DSP / standards strategy	Practical scalable implementation	Si-photonics + hybrid DFB, external modulation, DSP/DBP, standards checks	YES

**Eye-safety is satisfactory only under the stated assignment screening assumption; final product certification requires the actual ANSI Z136 accessible-emission calculation.*

Final Engineering Decision & Deployment Recommendation

FINAL DEPLOYMENT DECISION: YES — WITH ENGINEERING CONTROLS

The proposed transmitter is suitable for the stated 8-channel DWDM design when the calculated constraints are explicitly managed. A 1550 nm DFB source provides the required spectral quality and high-speed operation. The calculated 8.4 mW output at 40 mA exceeds the 2 mW/channel launch requirement, while the 20 GHz modulation bandwidth provides a 2.5 GHz margin over the 17.5 GHz NRZ requirement.

The dominant practical weakness is wavelength drift: an uncontrolled 40 °C temperature excursion would shift the wavelength by 4.0 nm. TEC stabilization to approximately ±1 °C, with fine current tuning, is therefore mandatory. External modulation is selected to reduce chirp and its interaction with dispersion.

The nonlinear analysis shows a total SPM phase shift of approximately 0.330 rad over six spans and an SBS threshold of approximately 1.59 mW under the stated coefficient assumptions. Since the 2 mW/channel launch is near/above that SBS threshold, the transmitter must use power equalization and SBS-aware spectral/phase management. SRS is not the limiting effect at 16 mW total launch. XPM and FWM remain relevant because eight channels share the fiber and the 100 GHz spacing allows nonlinear products and inter-channel phase coupling.

Soliton transmission is rejected because N = 0.171 and the fundamental-soliton power would be about 68.4 mW, far above the specified 2 mW/channel launch. DSP-based compensation is therefore the preferred method for residual dispersion and Kerr effects, with complexity reduced through practical per-channel/reduced-bandwidth approaches rather than full-bandwidth eight-channel DBP by default.

For implementation, Si-photonics with a hybrid-integrated 1550 nm DFB laser is recommended because it offers a compact and scalable platform for eight channels and allows integration of modulators, multiplexing, monitoring and control. The final product should undergo optical-interface checks against applicable ITU-T G.957 requirements, relevant IEEE 802.3bs/802.3bm PHY requirements where those interfaces are used, and a complete ANSI Z136 eye-safety assessment.

Conclusion

The engineering solution meets the assignment objectives by connecting the communication requirement to quantitative source selection, modulation design, thermal wavelength stabilization, nonlinear analysis, soliton feasibility, DSP compensation and photonic integration. The final recommended transmitter is an eight-channel, 1550 nm DFB-laser/externally-modulated Si-photonics architecture with TEC wavelength locking, controlled 2 mW/channel launch power, SBS-aware spectral management, optical filtering/equalization and practical DSP nonlinear/dispersion compensation. The design should be deployed only after final interface, thermal, optical-power and ANSI Z136 product-level verification.